



Modelling Financial Markets

Seminar 8: Backtesting and Evaluating Risk Forecasts — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ ACI: Adaptive Conformal Inference
- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CLT: Central Limit Theorem
- ▣ DM: Diebold–Mariano (test)
- ▣ DQ: Dynamic Quantile (test)
- ▣ ES: Expected Shortfall
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HS: Historical Simulation
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ PIT: Probability Integral Transform
- ▣ POF: Proportion Of Failures (test)

Outline

Introduction

Key Formulas

Part A: Derivations

Part B: Computation and Inference

Part C: Open Questions, AI Critique and a Case Study

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the asymptotics, size and power of VaR backtests and the consistency of scoring functions
- ▣ Backtest ES with residuals and the Acerbi–Szekely Z_2 by simulation
- ▣ Rank (VaR, ES) forecasts with the FZ0 loss, Diebold–Mariano tests and the MCS
- ▣ Build a conformal VaR and check its coverage in calm and crisis periods

- ▣ Background: Lecture 8; textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** ten derivations (A1–A10)
- ▣ **Part B:** nine computational exercises with inference (B1–B9)
- ▣ **Part C:** an open question, a seed for the team project (C1), the critique of an AI answer (C2) and a research case study (C3)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Report every test with its p -value and say how it was obtained (asymptotic, exact, bootstrap, simulation)
- A model that passes a test is not validated: ask how much power the test had

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (POF asymptotics and power), A3 (Markov-chain likelihood), A5 (traffic light), A6 (consistency of pinball and FZ0), A7 ($E[Z_2] = 0$), A8 (conformal quantile, ACI)
- Part B: B1 (VaR tests, DQ and size, S&P 500), B3 (ES tests with simulation), B5 (Diebold–Mariano with HAC, Giacomini–White), B7 (conformal VaR in crises)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A9 (model: A6), A10 (model: A7)
- Part B: B2 (model: A5, B1), B4 (model: B3), B6 (model: B5), B8 (model: B7), B9 (model: B1)
- Part C: C1, open question and project seed; C2, critique of an AI answer; C3, precision audit of ES rankings (model: B5)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter8_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `arch`, `matplotlib`
- Series: S&P 500, BET, Bitcoin (daily closes); EUR/RON: BNR reference rate; up to 18 September 2026
- Forecasts: the six models of Lecture 8 (HS, Normal, Student-t, GARCH-t, FHS, GARCH-EVT), rolling window of 1000 days, parameters re-estimated every 20 days
- Losses: $L_t = -r_t$ with $r_t = \ln P_t - \ln P_{t-1}$; breach: $I_t = 1\{L_t > \text{VaR}_t\}$

Key Formulas (1/2): VaR Backtests

- ▣ Kupiec [Kupiec \(1995\)](#): $LR_{uc} = -2 [(T - x) \ln(1 - p) + x \ln p - (T - x) \ln(1 - \hat{\pi}) - x \ln \hat{\pi}]$, $\hat{\pi} = x/T$, χ_1^2
- ▣ Christoffersen [Christoffersen \(1998\)](#): $\hat{\pi}_{01} = \frac{n_{01}}{n_{00} + n_{01}}$, $\hat{\pi}_{11} = \frac{n_{11}}{n_{10} + n_{11}}$, $\hat{\pi} = \frac{n_{01} + n_{11}}{n}$
- ▣ $LR_{ind} = -2 [(n_{00} + n_{10}) \ln(1 - \hat{\pi}) + (n_{01} + n_{11}) \ln \hat{\pi} - n_{00} \ln(1 - \hat{\pi}_{01}) - n_{01} \ln \hat{\pi}_{01} - n_{10} \ln(1 - \hat{\pi}_{11}) - n_{11} \ln \hat{\pi}_{11}]$
- ▣ $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi_2^2$; durations [Christoffersen & Pelletier \(2004\)](#): continuous Weibull approximation, H_0 : shape $b = 1$ (memoryless), scale free
- ▣ DQ [Engle & Manganelli \(2004\)](#): $\text{Hit}' X (X' X)^{-1} X' \text{Hit} / (\alpha(1 - \alpha)) \sim \chi_k^2$, $\text{Hit}_t = I_t - \alpha$, $X_t \in \mathcal{F}_{t-1}$
- ▣ Traffic light [BCBS \(1996\)](#): $X \sim \text{Binomial}(250, 0.01)$; yellow from $P(X \leq x) \geq 95\%$, red from $P(X \leq x) \geq 99.99\%$

Key Formulas (2/2): ES, Scores and Conformal VaR

- McNeil–Frey [McNeil & Frey \(2000\)](#): $e_t = (L_t - \text{ES}_t)/\sigma_t$ on breach days; $H_0: E[e_t] = 0$, $H_1: E[e_t] > 0$
- Acerbi–Szekely [Acerbi & Szekely \(2014\)](#): $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t l_t}{\text{ES}_t}$, $Z_2 = 1 - \frac{1}{T\alpha} \sum_t \frac{L_t l_t}{\text{ES}_t}$; fixed thresholds -0.70 and -1.8 for $T = 250$
- FZ0 [Patton, Ziegel & Chen \(2019\)](#): $\frac{\mathbf{1}_{\{L > v\}}(L - v)}{\alpha e} + \frac{v}{e} + \ln e - 1$, with VaR v and ES e (positive losses)
- Diebold–Mariano [Diebold & Mariano \(1995\)](#): $\bar{d}/\sqrt{\widehat{\text{LRV}}/T}$ with Newey–West weights [Newey & West \(1987\)](#); MCS [Hansen, Lunde & Nason \(2011\)](#); Giacomini–White [Giacomini & White \(2006\)](#): $T\bar{Z}'\hat{\Omega}^{-1}\bar{Z} \sim \chi_q^2$, $Z_t = h_{t-1}d_t$
- Conformal VaR: $\hat{\sigma}_t s_{(k)}$, $k = \lceil (n+1)(1-\alpha) \rceil$; ACI [Gibbs & Candès \(2021\)](#): $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$

A1: Asymptotics and Power of the POF Test [Solved]

Task

- **Context:** no data; a VaR 1% forecast is evaluated on T days, and the hits $I_t = 1\{L_t > \text{VaR}_t\}$ are i.i.d. Bernoulli(π).
 - ▶ Notation: $x = \sum_t I_t$, $\hat{\pi} = x/T$, log-likelihood $\ell(\pi) = x \ln \pi + (T - x) \ln(1 - \pi)$, Kupiec statistic $\text{LR}_{uc} = 2[\ell(\hat{\pi}) - \ell(\alpha)]$ Kupiec (1995).
- (a) Expand $\ell(\alpha)$ to second order around $\hat{\pi}$ and show that, under $H_0: \pi = \alpha$, LR_{uc} converges in distribution to χ_1^2 .
- (b) Let the true probability be the local alternative $\pi_T = \alpha + h/\sqrt{T}$; derive the limit of LR_{uc} and give its noncentrality parameter λ .
- (c) For $T = 250$, $\alpha = 1\%$ and $\pi = 2\%$, compute the local power $P(\chi_1^2(\lambda) > 3.84)$, the exact Binomial power of the 5% test and its exact size.
- **Report** λ , the two powers and the size, and **answer:** can one year of VaR 1% data detect a model that doubles the breach rate?

A1: Asymptotics and Power of the POF Test: Solution (1/2) [Solved]

Solution

- (a) Step 1: $\ell'(\hat{\pi}) = x/\hat{\pi} - (T - x)/(1 - \hat{\pi}) = 0$ at the MLE, so the first-order term of the expansion vanishes.
- Step 2: $\ell(\alpha) = \ell(\hat{\pi}) + \frac{1}{2}\ell''(\bar{\pi})(\alpha - \hat{\pi})^2$ with $\bar{\pi}$ between α and $\hat{\pi}$, and $-\ell''(\bar{\pi})/T = \frac{\hat{\pi}}{\bar{\pi}^2} + \frac{1-\hat{\pi}}{(1-\bar{\pi})^2} \xrightarrow{P} \frac{1}{\alpha(1-\alpha)}$.
- Step 3: therefore $\text{LR}_{uc} = \frac{T(\hat{\pi} - \alpha)^2}{\alpha(1 - \alpha)} + o_p(1) = Z_T^2 + o_p(1)$ with $Z_T = \frac{\sqrt{T}(\hat{\pi} - \alpha)}{\sqrt{\alpha(1 - \alpha)}}$.
- Step 4: by the CLT for i.i.d. Bernoulli variables $Z_T \xrightarrow{d} N(0, 1)$, hence $\text{LR}_{uc} \xrightarrow{d} \chi_1^2$; the 5% critical value is 3.84.
- (b) Under π_T : $\sqrt{T}(\hat{\pi} - \alpha) = \sqrt{T}(\hat{\pi} - \pi_T) + h$, so $Z_T \xrightarrow{d} N(h/\sqrt{\alpha(1 - \alpha)}, 1)$ and $\text{LR}_{uc} \xrightarrow{d} \chi_1^2(\lambda)$ with

$$\lambda = \frac{h^2}{\alpha(1 - \alpha)} = \frac{T(\pi - \alpha)^2}{\alpha(1 - \alpha)}.$$

A1: Asymptotics and Power of the POF Test: Solution (2/2) [Solved]

Solution

- ▣ (c) Noncentrality: $\lambda = \frac{250(0.02 - 0.01)^2}{0.01 \cdot 0.99} = \frac{0.025}{0.0099} = 2.53$; local power $P(\chi_1^2(2.53) > 3.84) = 36\%$.
- ▣ Exact evaluation of the asymptotic LR rule: $\text{LR}_{uc}(x) \leq 3.84$ exactly for $x \in \{1, \dots, 6\}$, so the acceptance region is $1 \leq x \leq 6$.
- ▣ Exact power: $1 - P(1 \leq X \leq 6 \mid X \sim \text{Bin}(250, 0.02)) = 1 - 75.7\% = 24\%$.
- ▣ Exact size: $P(X = 0) + P(X \geq 7)$ under $\text{Bin}(250, 0.01) = 8.11\% + 1.37\% = 9.5\%$, almost twice the nominal 5%.
- ▣ Interpretation: with $\alpha T = 2.5$ expected breaches the lattice of x dominates; a model that doubles the breach rate passes one year of testing about three times out of four.

A2: Power of the POF Test at the FRTB Desk Level [Proposed]

Solution discussed in class

Task

- **Context:** no data; a trading desk reports VaR 2.5% ($\alpha = 0.025$, the FRTB desk level) for one year, $T = 250$; hits are i.i.d. Bernoulli(π).
- (a) Compute the noncentrality λ and the local power of the 5% Kupiec test against $\pi = 5\%$; then find the exact acceptance region of x and compute the exact Binomial power and size.
- (b) Find λ^* with $P(\chi_1^2(\lambda^*) > 3.84) = 0.80$ and solve $T(\pi - \alpha)^2 / (\alpha(1 - \alpha)) = \lambda^*$ for the number of days T ; then find the first T at which the exact power reaches 80%.
- (c) Compute $P(X > 30)$ for $X \sim \text{Bin}(250, 0.05)$, the probability that a desk with a doubled breach rate exceeds the FRTB limit of 30 exceptions.
- **Report** λ , the powers, the size, the two values of T and $P(X > 30)$; **answer:** what does the FRTB desk limit actually screen for?
- Model: A1

A3: Christoffersen Test and the Markov-Chain Likelihood [Solved]

Task

- ▣ **Context:** BET index (data/market/BET.csv), daily VaR 1% of historical simulation (window 1000 days), 2007–2010, 1005 days.
 - ▶ Transition counts of the hit sequence: $n_{00} = 958$, $n_{01} = 22$, $n_{10} = 22$, $n_{11} = 2$ (n_{ij} : days in state j after a day in state i); $LR_{uc} = 14.11$ is given.
- ▣ (a) Compute the transition probabilities $\hat{\pi}_{01} = n_{01}/(n_{00} + n_{01})$, $\hat{\pi}_{11} = n_{11}/(n_{10} + n_{11})$ and the pooled $\hat{\pi} = (n_{01} + n_{11})/n$.
- ▣ (b) Compute the independence statistic LR_{ind} Christoffersen (1998), its χ^2_1 p -value and the decision at 5%.
- ▣ (c) Compute $LR_{cc} = LR_{uc} + LR_{ind}$, its χ^2_2 p -value and the decision at 5%.
- ▣ (d) Write the likelihood of the first-order Markov chain conditional on I_1 , derive $\hat{\pi}_{01}$ and $\hat{\pi}_{11}$ as MLEs, and explain why LR_{cc} has 2 degrees of freedom.
- ▣ **Report** the three probabilities, both statistics with p -values, and **answer**: does HS fail because of too many breaches or because breaches cluster?

A3: Christoffersen Test and the Markov-Chain Likelihood: Solution (1/2) [Solved]

Solution

- (a) $\hat{\pi}_{01} = 22/(958 + 22) = 0.0224$; $\hat{\pi}_{11} = 2/(22 + 2) = 0.0833$; $\hat{\pi} = (22 + 2)/1004 = 0.0239$.
- After a breach the breach probability is almost four times higher (0.0833 versus 0.0224).
- (b) $LR_{ind} = -2[\ln L(\hat{\pi}, \hat{\pi}) - \ln L(\hat{\pi}_{01}, \hat{\pi}_{11})] = 2.32$, $p = 0.128 > 0.05$: independence is not rejected, because $\hat{\pi}_{11}$ rests on only 2 consecutive breaches.
- (c) $LR_{cc} = 14.11 + 2.32 = 16.43$, $p = 0.0003 < 0.05$: conditional coverage is rejected, driven by the coverage part.
- Interpretation: HS fails mainly through too many breaches ($\hat{\pi} \approx 2.4\%$ instead of 1%); the clustering is visible but the test lacks the data to confirm it.

A3: Christoffersen Test and the Markov-Chain Likelihood: Solution (2/2) [Solved]

Solution

- (d) Step 1: conditional on I_1 , $\ln L(\pi_{01}, \pi_{11}) = n_{00} \ln(1 - \pi_{01}) + n_{01} \ln \pi_{01} + n_{10} \ln(1 - \pi_{11}) + n_{11} \ln \pi_{11}$.
- Step 2: the log-likelihood separates into a part in π_{01} and a part in π_{11} ; setting $\partial/\partial\pi_{i1} = -n_{i0}/(1 - \pi_{i1}) + n_{i1}/\pi_{i1} = 0$ gives $\hat{\pi}_{i1} = n_{i1}/(n_{i0} + n_{i1})$, two Bernoulli MLEs.
- Step 3: $LR_{cc} = -2[\ln L(\alpha, \alpha) - \ln L(\hat{\pi}_{01}, \hat{\pi}_{11})]$ fixes both free parameters, $\pi_{01} = \alpha$ and $\pi_{11} = \alpha$: two restrictions, hence χ^2_2 .
- Step 4: adding and subtracting $\ln L(\hat{\pi}, \hat{\pi})$ splits it exactly into $LR_{uc} + LR_{ind}$ when both are computed on the same $T - 1$ transitions; if LR_{uc} uses all T days, the first observation enters only one term and the identity holds only asymptotically.

A4: The DQ Statistic as a Wald Test [Proposed]

Solution discussed in class

Task

- **Context:** no data; $\text{Hit}_t = I_t - \alpha$ and k regressors X_t known at $t - 1$ (constant, lagged hits, the VaR level); linear probability model $\text{Hit}_t = X_t\beta + u_t$ [Engle & Manganelli \(2004\)](#).
 - ▶ Null hypothesis: correct conditional coverage, $P(I_t = 1 \mid \mathcal{F}_{t-1}) = \alpha$.
- (a) Show that under H_0 $E[\text{Hit}_t \mid \mathcal{F}_{t-1}] = 0$ and $\text{Var}(\text{Hit}_t \mid \mathcal{F}_{t-1}) = \alpha(1 - \alpha)$.
- (b) Using a martingale-difference CLT for $T^{-1/2}X'\text{Hit}$, derive $\text{DQ} = \text{Hit}'X(X'X)^{-1}X'\text{Hit}/(\alpha(1 - \alpha)) \xrightarrow{d} \chi_k^2$.
- (c) Take $X_t = 1$ only: show that DQ becomes $T(\hat{\pi} - \alpha)^2/(\alpha(1 - \alpha))$, compute it for $T = 250$, $x = 7$, $\alpha = 1\%$ with its χ_1^2 p -value, and compare with $\text{LR}_{uc} = 5.50$ ($p = 0.019$).
- **Report** the derivation and the two statistics; **answer:** why do two asymptotically equivalent tests disagree here?
- Model: A3

A5: The Traffic Light from the Binomial Distribution [Solved]

Task

- ▣ **Context:** no data; the Basel traffic light [BCBS \(1996\)](#) counts the exceptions X of a VaR 1% over the last 250 days; for a correct model $X \sim \text{Binomial}(250, 0.01)$.
- ▣ (a) Compute $P(X \leq x)$ for $x = 4, 5, 9, 10$ and derive the zones: green while $P(X \leq x) < 95\%$, red from $P(X \leq x) \geq 99.99\%$, yellow in between.
- ▣ (b) A bank reports 6 exceptions: give the zone, the 1996 plus factor, the FRTB capital multiplier and the probability $P(X \geq 6)$ for a correct model.
- ▣ (c) Compute the probability that a model with a true breach probability of 3% still lands in the green zone, $P(X \leq 4 \mid p = 0.03)$.
- ▣ **Report** the four probabilities, the zones and the answers to (b) and (c); **answer:** which error does the traffic light control, and which does it leave large?

A5: The Traffic Light from the Binomial Distribution: Solution [Solved]

Solution

- ▣ (a) $P(X \leq 4) = 89.22\%$, $P(X \leq 5) = 95.88\%$, $P(X \leq 9) = 99.97\%$, $P(X \leq 10) = 99.99\%$.
- ▣ The 95% threshold is first crossed at $x = 5$ and 99.99% at $x = 10$: green 0–4, yellow 5–9, red 10 or more.
- ▣ (b) 6 exceptions: yellow zone; 1996 plus factor 0.50 (multiplier 3.50); FRTB multiplier 1.76;
 $P(X \geq 6) = 1 - P(X \leq 5) = 4.12\%$: a correct model rarely produces this many.
- ▣ (c) $P(X \leq 4 \mid p = 0.03) = 12.8\%$: a model that triples the breach rate stays green about one time in eight.
- ▣ Interpretation: a correct model leaves green with probability $P(X \geq 5) = 10.78\%$ (about one year in nine) and almost never reaches red: the zones control false alarms of the red zone, while the type II error against bad models remains large with 250 days.

A6: Consistency of the Pinball and FZ0 Losses [Solved]

Task

- ▣ **Context:** a loss L with continuous cdf F and density f ; tail probability $\alpha \in (0, 1)$; forecasts v (VaR) and e (ES), with $e \geq v > 0$.
- ▣ (a) Differentiate the expected pinball loss $E[(1\{L > v\} - \alpha)(L - v)]$ in v and show that its minimiser is VaR_α .
- ▣ (b) For $\text{FZ0}(v, e; L) = \frac{1\{L > v\}(L - v)}{\alpha e} + \frac{v}{e} + \ln e - 1$ [Patton, Ziegel & Chen \(2019\)](#), compute $\partial_v E[\text{FZ0}]$ and $\partial_e E[\text{FZ0}]$, show that both vanish at $(\text{VaR}_\alpha, \text{ES}_\alpha)$ and write the two identification functions.
- ▣ (c) Check numerically: simulate 10^6 Student- t_4 losses with unit variance, minimise the average pinball loss and the average FZ0 loss at $\alpha = 2.5\%$, and compare with the true VaR and ES.
- ▣ **Report** the derivatives, the identification functions and the numerical minimisers; **answer:** why does consistency make ranking by average score meaningful?

A6: Consistency of the Pinball and FZ0 Losses: Solution [Solved]

Solution

- (a) $\frac{d}{dv} E[(1\{L > v\} - \alpha)(L - v)] = -P(L > v) + \alpha = F(v) - (1 - \alpha)$; setting it to zero gives $F(v) = 1 - \alpha$, i.e. $v = \text{VaR}_\alpha$; the second derivative $f(v) > 0$ confirms a minimum.
- (b) $\partial_v E[\text{FZ0}] = \frac{1}{e} \left(1 - \frac{P(L > v)}{\alpha} \right)$, zero exactly when $P(L > v) = \alpha$, i.e. $v = \text{VaR}_\alpha$.
- $\partial_e E[\text{FZ0}] = \frac{1}{e^2} \left(e - v - \frac{E[(L - v)1\{L > v\}]}{\alpha} \right)$; at $v = \text{VaR}_\alpha$, $E[(L - v)1\{L > v\}]/\alpha = E[L - v \mid L > v]$, so $e = E[L \mid L > v] = \text{ES}_\alpha$.
- Identification functions: $V_1(v, L) = \alpha - 1\{L > v\}$ and $V_2(v, e, L) = e - v - \frac{1}{\alpha} 1\{L > v\}(L - v)$; their expectations vanish only at the true pair (Fissler & Ziegel (2016)).
- (c) Numerical minimisers: pinball $v = 1.960$; FZ0 $(v, e) = (1.960, 2.812)$; truth $(1.963, 2.824)$: equal up to simulation error.
- Interpretation: a forecaster who knows F minimises the expected score by reporting the truth, so a lower average score is evidence of a better forecast.

A7: Why $E[Z_2] = 0$, and Z_1, Z_2 on One Year [Solved]

Task

- ▣ **Context:** ES 2.5% forecasts ($\alpha = 0.025$) evaluated with the Acerbi–Szekely statistics Acerbi & Szekely (2014);
 $l_t = 1\{L_t > \text{VaR}_t\}$, $N_T = \sum_t l_t$.
▶ $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t l_t}{\text{ES}_t}$, $Z_2 = 1 - \frac{1}{T\alpha} \sum_t \frac{L_t l_t}{\text{ES}_t}$
- ▣ (a) Assuming a continuous and correct forecast distribution F_t , prove $E[L_t l_t \mid \mathcal{F}_{t-1}] = \alpha \text{ES}_t$ and deduce $E[Z_2] = 0$; explain on what Z_1 conditions.
- ▣ (b) One year, $T = 250$, 8 breach days with losses (%) 3.9, 2.8, 4.6, 3.1, 5.2, 2.9, 3.6, 7.4 and ES forecasts (%) 3.4, 3.3, 3.6, 3.2, 3.5, 3.3, 3.4, 3.8: compute the ratios L_t/ES_t , Z_1 and Z_2 , and classify Z_2 with the thresholds -0.70 (yellow) and -1.8 (red).
- ▣ (c) Recompute Z_2 after adding one more breach with $L = 7\%$ and $\text{ES} = 3.5\%$.
- ▣ **Report** the proof, Z_1 , Z_2 and the zone; **answer:** which days drive Z_2 ?

A7: Why $E[Z_2] = 0$, and Z_1, Z_2 on One Year: Solution [Solved]

Solution

- (a) $E[L_t 1\{L_t > \text{VaR}_t\} \mid \mathcal{F}_{t-1}] = P(L_t > \text{VaR}_t \mid \mathcal{F}_{t-1}) E[L_t \mid L_t > \text{VaR}_t, \mathcal{F}_{t-1}] = \alpha \text{ES}_t$.
- ES_t is known at $t - 1$, so $E[L_t / \text{ES}_t \mid \mathcal{F}_{t-1}] = \alpha$ for every t and $E[Z_2] = 1 - T\alpha / (T\alpha) = 0$: the denominator $T\alpha$ is a constant.
- Z_1 divides by the random N_T : it has mean zero only conditionally on $N_T \geq 1$, is undefined when $N_T = 0$, and ignores how many breaches occurred.
- (b) Ratios: 1.147, 0.848, 1.278, 0.969, 1.486, 0.879, 1.059, 1.947; their sum is 9.613.
- $Z_1 = 1 - 9.613/8 = -0.202$: tail losses are on average 20% above the forecast ES ;
 $Z_2 = 1 - 9.613/6.25 = -0.538 > -0.70$: green zone.
- (c) With one more ratio $7/3.5 = 2$: $Z_2 = -0.86 < -0.70$, yellow. Interpretation: Z_2 reacts to both the number and the size of breaches, and a single large loss can change the zone.

A8: Conformal Quantile and One ACI Step [Solved]

Task

- ▣ **Context:** $n = 19$ calibration scores $s = L/\hat{\sigma}$ (sorted): 0.19, 0.35, 0.52, 0.61, 0.73, 0.88, 0.94, 1.05, 1.12, 1.28, 1.47, 1.66, 1.83, 1.95, 2.02, 2.28, 2.41, 2.75, 3.10; tomorrow's volatility forecast is $\hat{\sigma} = 1.4\%$.
- ▣ (a) Compute the split conformal VaR 10%: the order statistic $s_{(k)}$ with $k = \lceil (n+1)(1-\alpha) \rceil$, times $\hat{\sigma}$.
- ▣ (b) Apply one ACI step [Gibbs & Candès \(2021\)](#) with $\gamma = 0.02$ and $\alpha_1 = 0.10$: compute $\alpha_2 = \alpha_1 + \gamma(\alpha - \text{err}_1)$ after a breach ($\text{err}_1 = 1$) and after no breach ($\text{err}_1 = 0$).
- ▣ (c) Recompute the order index and the VaR for both values of α_2 .
- ▣ **Report** k , the three VaR values and **answer**: how smoothly does ACI move the VaR?

A8: Conformal Quantile and One ACI Step: Solution [Solved]

Solution

- (a) $k = \lceil 20 \cdot 0.9 \rceil = 18$; $s_{(18)} = 2.75$; $\text{VaR} = 1.4 \cdot 2.75 = 3.85\%$.
- (b) After a breach: $\alpha_2 = 0.10 + 0.02(0.10 - 1) = 0.082$; after no breach: $\alpha_2 = 0.10 + 0.02 \cdot 0.10 = 0.102$.
- (c) After a breach: $k = \lceil 20(1 - 0.082) \rceil = \lceil 18.36 \rceil = 19$, $\text{VaR} = 1.4 \cdot 3.10 = 4.34\%$, a higher VaR.
- After no breach: $k = \lceil 17.96 \rceil = 18$, the same index, so the VaR does not move.
- Interpretation: with small n ACI moves in jumps of one order statistic; small changes of α_t are invisible until they cross a grid point $j/(n+1)$.

A9: ES Alone Is Not Elicitable [Proposed]

Solution discussed in class

Task

- **Context:** a functional $T(F)$ is elicitable if $T(F) = \arg \min_x E_F[S(x, L)]$ for some scoring function S [Gneiting \(2011\)](#); here $T = \text{ES}_\alpha$ with $\alpha < 1/2$.
- (a) Prove that if T is elicitable, every level set $\{F : T(F) = t\}$ is convex, i.e. closed under mixtures $F_\lambda = (1 - \lambda)F_0 + \lambda F_1$.
- (b) Let $F_0 = \delta_1$ (loss 1 with certainty) and F_1 with $P(L = 2) = \alpha/2$, $P(L = 0) = 1 - \alpha/2$; compute ES_α of F_0 , of F_1 and of the mixture $\frac{1}{2}F_0 + \frac{1}{2}F_1$ from the upper α tail mass.
- (c) Compute VaR_α of F_0 and F_1 and explain why the example does not contradict the joint elicibility of (VaR, ES) .
- **Report** the three ES values; **answer:** what does this imply for ranking ES forecasts alone?
- Model: A6

A10: Null Moments of the Cumulative Violation [Proposed]

Solution discussed in class

Task

- ▣ **Context:** under a correct forecast distribution the PITs $u_t = F_t(L_t)$ are i.i.d. $U(0, 1)$; the cumulative violation of Du & Escanciano (2017) is $H_t = \frac{1}{\alpha}(u_t - (1 - \alpha))1\{u_t > 1 - \alpha\}$, with $\alpha = 2.5\%$.
- ▣ (a) Compute $E[H_t]$ and $E[H_t^2]$ by integrating over $u \in (1 - \alpha, 1)$, and deduce $\text{Var}(H_t) = \alpha(1/3 - \alpha/4)$.
- ▣ (b) Check both moments by simulating $4 \cdot 10^6$ uniform draws.
- ▣ (c) Compute the standard error of $\bar{H}$ for $T = 250$ and the value above which $\bar{H}$ rejects H_0 at 5% (one-sided, Normal approximation).
- ▣ **Report** the moments, the simulation check and the critical value; **answer:** why does H_t carry more information than I_t ?
- ▣ Model: A7

B1: Backtesting Six VaR Models on the S&P 500 (1/2) [Solved]

Task

- **Context:** S&P 500 daily closes (`data/market/GSPC.INDX.csv`), losses $L_t = -\ln(P_t/P_{t-1})$; one-day forecasts of the six models of Lecture 8 (HS, Normal, Student-t, GARCH-t, FHS, GARCH-EVT) on a rolling 1000-day window, parameters re-estimated every 20 days; evaluation from January 2007 to 18.09.2026, $T = 4\,959$ days.
- (a) For each model count the breaches of VaR 1% and compute the Kupiec p -value, both asymptotic (χ^2_1) and exact (two-sided Binomial).
- (b) Compute the Christoffersen independence and conditional coverage p -values [Christoffersen \(1998\)](#).
- (c) Fit the Weibull duration model [Christoffersen & Pelletier \(2004\)](#) with censored first and last durations; report the shape b and the LR p -value for $H_0: b = 1$.

B1: Backtesting Six VaR Models on the S&P 500 (2/2) [Solved]

Task

- (d) Run the DQ test [Engle & Manganelli \(2004\)](#): regress Hit_t on a constant, four lagged hits and VaR_t and compare DQ with χ^2_6 .
- (e) Estimate the size of the battery: simulate 1000 paths from a GARCH(1,1)-t fitted to the S&P 500 (1990–2026), backtest the true VaR 1% at $T = 250$ and $T = 5000$, and record the rejection rates at 5%.
- **Report** one table (breaches, rate, five p -values per model) and the simulated sizes.
- **Answer:**
 - ▶ Which models pass all tests, and which test is decisive?
 - ▶ Why can a model with the right number of breaches still fail?
 - ▶ Given the simulated size, how would you read a DQ p -value of 0.03 at $T = 250$?

B1: Backtesting Six VaR Models on the S&P 500: Solution (1/2) [Solved]

Model	x	Rate (%)	p_{uc}	p_{exact}	p_{ind}	p_{cc}	b / p_{dur}	p_{DQ}
HS	83	1.67	<0.001	<0.001	<0.001	<0.001	0.52 / <0.001	<0.001
Normal	134	2.70	<0.001	<0.001	<0.001	<0.001	0.56 / <0.001	<0.001
Student-t	99	2.00	<0.001	<0.001	<0.001	<0.001	0.53 / <0.001	<0.001
GARCH-t	88	1.77	<0.001	<0.001	0.095	<0.001	0.94 / 0.479	<0.001
FHS	61	1.23	0.116	0.127	0.046	0.040	0.93 / 0.514	<0.001
GARCH-EVT	56	1.13	0.370	0.395	0.028	0.060	0.88 / 0.240	<0.001

- Procedure: build the hit series I_t for each model, then apply the five tests to the same series; all decisions use the 5% significance level.
- (a)–(c): only GARCH-EVT passes Kupiec, conditional coverage and durations; FHS fails p_{cc} narrowly; HS, Normal and Student-t fail the duration test with $b \approx 0.5$ (clustered breaches).
- With 4959 days asymptotic and exact Kupiec p -values agree; in 2008 alone ($T = 253$, $x = 8$) they are 0.0058 and 0.0086.

B1: Backtesting Six VaR Models on the S&P 500: Solution (2/2) [Solved]

Solution

- (d) DQ rejects all six models, including GARCH-EVT ($p < 0.001$): hits are predictable from lagged hits and from the VaR level, a conditional failure invisible to the counts.
- (e) Simulated size at 5%, $T = 250$: POF 9.8%, CC 1.2%, durations 14.8%, DQ 8.6%; $T = 5000$: POF 4.9%, CC 3.4%, durations 4.6%, DQ 8.8%.
- Reading: in one year the tests are mis-sized (POF too often, CC too rarely); DQ stays oversized even at 5000 days, so a DQ p -value of 0.03 at $T = 250$ is weak evidence and needs a simulated critical value.
- Economic conclusion: a model with the right number of breaches (FHS) can still fail on timing; only filtered models with a flexible tail survive most tests.

B2: Traffic Light for GARCH-t on the BET [Proposed]

Solution discussed in class

Task

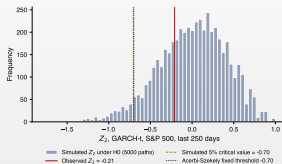
- ▣ **Context:** BET index (data/market/BET.csv), VaR 1% and VaR 2.5% of GARCH-t (rolling 1000-day window, refit every 20 days), 2007 to 18.09.2026; Basel zones from A5.
- ▣ (a) Count the VaR 1% exceptions in the last 250 days; give the zone, the cumulative Binomial probability $P(X \leq x)$ and the FRTB multiplier.
- ▣ (b) Count exceptions in every rolling 250-day window since 2007; report the share of days in each zone and the worst window with its end date.
- ▣ (c) Count the VaR 2.5% exceptions in the last 250 days and compare with the FRTB desk limit of 30.
- ▣ **Answer:** would a supervisor have penalised this model since 2007? Is time in the yellow zone evidence against the model or bad luck (use A5)? What does the traffic light ignore that Z_2 would capture?
- ▣ Model: A5 (zones), B1 (breach series)

B3: ES Backtests with Bootstrap and Simulation [Solved]

Task

- **Context:** S&P 500 (GSPC.INDX.csv), forecasts of (VaR 2.5%, ES 2.5%) of GARCH-t, FHS and HS, 2007 to 18.09.2026, $T = 4\,959$ days (setting of B1).
- (a) On breach days compute the exceedance residuals $e_t = (L_t - \text{ES}_t)/\sigma_t$ [McNeil & Frey \(2000\)](#); report their number, mean and one-sided t -statistic, and compare the bootstrap p -value (5000 resamples of the centred residuals) with the Student- t p -value.
- (b) Compute Z_2 [Acerbi & Szekely \(2014\)](#) on the full sample and on the last 250 days; obtain p -values and 5% critical values from 5000 paths simulated from each model's forecast distributions.
- **Answer:** why is the bootstrap preferred for the residuals? Does the simulated 5% critical value for 250 days agree with the fixed -0.70 ? Why is the full-sample critical value much closer to zero?

B3: ES Backtests with Bootstrap and Simulation: Solution (1/2) [Solved]



Model	n	$\bar{e} / t$	p_{boot} / p_t	$Z_2 (T = 4959)$	$p / \text{crit.}$	$Z_2 (250 \text{ d}), p$
GARCH-t	192	0.106 / 1.80	0.023 / 0.037	-0.618	<0.001 / -0.153	-0.210, 0.289
FHS	130	0.093 / 1.21	0.096 / 0.114	-0.091	0.154 / -0.150	0.142, 0.602
HS	156	0.363 / 2.88	<0.001 / 0.002	-0.405	<0.001 / -0.154	0.723, 0.975

- Procedure: (a) centre the residuals, resample them 5000 times and read the share of bootstrap means above the observed mean; (b) simulate $L_t^* = F_t^{-1}(U_t)$ day by day and recompute Z_2^* .

B3: ES Backtests with Bootstrap and Simulation: Solution (2/2) [Solved]

Solution

- (a) GARCH-t: $\bar{e} = 0.106$, $t = 1.80$, bootstrap $p = 0.023$ versus Student- t $p = 0.037$: the residuals are skewed, so the t approximation is poor and the bootstrap (i.i.d. resampling, which assumes independent residuals) is preferred.
- (b) Full sample: GARCH-t ($Z_2 = -0.618$, $p < 0.001$) and HS are rejected at 5%; FHS passes ($p = 0.154$).
- Last 250 days: the simulated 5% critical value for GARCH-t is -0.705 , close to the fixed -0.70 ; with only 6 breaches no model is rejected.
- Reading: the null distribution of Z_2 shrinks like $1/\sqrt{T\alpha}$, so the full-sample critical value (-0.153) is much closer to zero; economically, GARCH-t underestimates the average tail loss, and FHS, with the same volatility filter, passes: the failure points to the t tail.

B4: Z_2 by Simulation on BET and Bitcoin [Proposed]

Solution discussed in class

Task

- ▣ **Context:** BET (BET.csv, from 2007) and Bitcoin (BTC-USD.CC.csv, from 2017) to 18.09.2026; (VaR 2.5%, ES 2.5%) forecasts of GARCH-t and GARCH-EVT (rolling 1000-day window).
- ▣ (a) Compute Z_2 on the full evaluation sample, with its p -value and 5% critical value from 5000 paths simulated from the forecast distributions.
- ▣ (b) Repeat on the last 250 days, report the number of breach days, and compare the critical value with the fixed -0.70 .
- ▣ **Answer:** which model would you trust for ES on the BET and on Bitcoin? Why does the verdict change between the full sample and the last year?
- ▣ Model: B3

B5: Diebold–Mariano with HAC Variance [Solved]

Task

- **Context:** S&P 500 (GSPC.INDX.csv), FZ0 losses [Patton, Ziegel & Chen \(2019\)](#) of the (VaR 2.5%, ES 2.5%) forecasts of the six models, 2007 to 18.09.2026 (setting of B1).
- (a) Compute the average FZ0 loss of each model and identify the best model.
- (b) Test GARCH-t, HS and GARCH-EVT against FHS with the DM statistic [Diebold & Mariano \(1995\)](#) and a Newey–West variance [Newey & West \(1987\)](#) with 9 lags; give the two-sided p -values.
- (c) Compute the naive t -statistic that ignores autocorrelation and the ratio of the long-run variance to the ordinary variance.
- (d) Run the Giacomini–White conditional test [Giacomini & White \(2006\)](#) with $h_{t-1} = (1, \Delta L_{t-1})$ and χ^2_2 critical values.
- **Answer:** which differences are significant? When does ignoring autocorrelation overstate significance? Does conditioning change the verdict for GARCH-EVT versus FHS?

B5: Diebold–Mariano with HAC Variance: Solution [Solved]

Solution

- (a) Average FZ0: FHS -3.551 , GARCH-EVT -3.548 , GARCH-t -3.512 , HS -3.134 , Student-t -3.036 , Normal -2.958 ; FHS is best.
- (b) $DM = \bar{d} / \sqrt{\widehat{LRV} / T}$. GARCH-t minus FHS: $\bar{d} = 0.0388$, $DM = 2.65$, $p = 0.008$: GARCH-t is significantly worse at 5%.
- HS minus FHS: $DM = 4.70$ ($p < 0.001$), significantly worse; GARCH-EVT minus FHS: $DM = 1.24$ ($p = 0.216$), no significant difference.
- (c) Naive t : GARCH-t 2.72, HS 7.18; for HS the long-run variance is 2.34 times the ordinary one, so the naive statistic is inflated by about $\sqrt{2.34}$: HS losses cluster in crises.
- (d) GW: GARCH-EVT 7.38 ($p = 0.025$), GARCH-t 7.59 ($p = 0.022$), HS 53.38 ($p < 0.001$); the DM tie of GARCH-EVT becomes a conditional rejection: yesterday's loss differential predicts which method wins today.
- Economic conclusion: the average FZ0 losses of FHS and GARCH-EVT are not significantly different, but their relative performance varies over time; the FZ0 gaps of GARCH-t and HS are statistically robust.

B6: Comparing Models on Bitcoin and EUR/RON [Proposed]

Solution discussed in class

Task

- ▣ **Context:** Bitcoin (BTC-USD.CC.csv, from 2017) and EUR/RON (BNR reference rate, from 2009) to 18.09.2026; FZ0 losses of the six models (setting of B5).
- ▣ (a) Find the best model in each market and test every other model against it with DM tests and Newey–West variance.
- ▣ (b) Compute the MCS p -values [Hansen, Lunde & Nason \(2011\)](#) with the T_{max} statistic, a block bootstrap (blocks of 10 days) and 1000 replications; list the 90% MCS.
- ▣ (c) On Bitcoin, VaR 2.5%, compute the mean elementary quantile scores [Ehm, Gneiting, Jordan & Krüger \(2016\)](#) of HS and GARCH-EVT on a grid of thresholds θ and check whether the Murphy curves cross.
- ▣ **Answer:** why is the MCS so large for Bitcoin and so small for EUR/RON? Relate it to the effective tail sample αT [Pele & Mazurencu-Marinescu-Pele \(2026\)](#).
- ▣ Model: B5

B7: Conformal VaR in Calm and Crisis Periods [Solved]

Task

- **Context:** S&P 500 (GSPC.INDX.csv), 2007 to 18.09.2026; score $s_t = L_t / \hat{\sigma}_t$ with $\hat{\sigma}_t$ the standard deviation of the previous 250 returns; calibration on the last 250 scores; crisis days: previous 20-day volatility above the 90th percentile of its values over the previous 1000 days (known in advance).
- (a) Build the split conformal VaR 5% and VaR 1% and compute the breach rates overall, on calm days and on crisis days, with the Kupiec p -value.
- (b) Repeat on simulated i.i.d. Student- t_4 data of the same length: the returns are exchangeable, the rolling-standardised scores only approximately so (their denominators overlap).
- (c) Apply the Christoffersen independence test to both breach series.
- **Answer:** is the marginal guarantee met on real data? What changes between real and exchangeable data: coverage or independence?

B7: Conformal VaR in Calm and Crisis Periods: Solution [Solved]

	All	Calm	Crisis	p_{uc}	p_{ind}	i.i.d.: all / calm / crisis
VaR 5%	5.34	5.02	7.29	0.272	<0.001	4.74 / 4.68 / 5.26
VaR 1%	1.29	1.15	2.10	0.049	0.001	0.73 / 0.72 / 0.78

- Procedure: for each day take the k -th order statistic of the last 250 scores, $k = \lceil 251(1 - \alpha) \rceil$, multiply by $\hat{\sigma}_t$, and record the hit; rates in %; crisis = 713 days.
- (a) VaR 5%: 5.34% overall (Kupiec $p = 0.272$, not rejected), but 7.29% on crisis days.
- (b)–(c) i.i.d. data: calm 4.68%, crisis 5.26%, independence holds ($p = 0.965$); real data: independence rejected ($p < 0.001$).
- Reading: unconditional coverage is not rejected for VaR 5% but is rejected for VaR 1% (Kupiec $p = 0.049$); non-rejection does not establish the conformal guarantee; volatility clustering breaks exchangeability, so breaches cluster and crisis coverage fails.

B8: Conformal VaR on BET and Bitcoin [Proposed]

Solution discussed in class

Task

- **Context:** BET (BET.csv) and Bitcoin (BTC-USD.CC.csv) to 18.09.2026; VaR 5% by split conformal, ACI with $\gamma = 0.005$, and GARCH-t; calm and crisis days defined as in B7.
- (a) Compute the breach rates overall, on calm days and on crisis days for the three methods in both markets.
- (b) Identify, for each market, the method closest to 5% on crisis days.
- **Answer:** does ACI fix conditional coverage or only the long-run average? Would you use a conformal VaR for regulatory capital?
- Model: B7

B9: Parametric Bootstrap Under Estimation Risk [Proposed]

Solution discussed in class

Task

- **Context:** data-generating process GARCH(1,1)-t with the parameters fitted to the S&P 500 (GSPC.INDX.csv, 1990–2026), as in B1(e); estimation window $R = 1000$ days, fixed scheme (estimate once, then forecast).
- (a) Simulate 400 paths of length $R + 5000$; on each, re-estimate GARCH-t on the first R days, forecast VaR 1% for the next P days, and run the Kupiec test for $P = 250, 1000$ and 5000 .
- (b) Compute the rejection rate at 5% with the estimated parameters and, on the same paths, with the true parameters.
- (c) Compute the standard deviation of the breach rate across paths and compare it with the Binomial value $\sqrt{\alpha(1 - \alpha)/P}$.
- **Answer:** how does the distortion grow with P/R (relate to [Escanciano & Olmo \(2010\)](#))? Which critical values would you use for $P/R \approx 5$?
- Model: B1 (part e)



C1: Which Risk Model Would You Certify for a Romanian Bank's Trading Book? (1/2) [Proposed]

Solution discussed in class

Task

- ▣ **Context:** a trading book in lei, half in the BET index (data/market/BET.csv) and half in euro valued at the BNR EUR/RON reference rate; daily data from 2005, evaluation from 09.07.2009 to 18.09.2026; the six models of Lecture 8 with a rolling 1000-day window.
- ▣ You are the model validator: recommend a VaR/ES model in this simplified exercise, combining illustrative regulatory checks with statistical evidence, and list what FRTB approval would also require (desk P&L backtests, P&L attribution).
- ▣ (a) Join the two price series on common days; rebalance daily to 50/50 value weights, without costs:
$$r_t^P = \ln \left(1 + \frac{1}{2} R_t^{BET} + \frac{1}{2} R_t^{EUR} \right),$$
 with R simple returns.
- ▣ (b) Forecast VaR 1%, VaR 2.5% and ES 2.5% with the six models and backtest them with at least three of: Kupiec, Christoffersen, durations, DQ, Z_2 .
- ▣ (c) Place each model in the traffic light for its worst and for its last 250-day window, and check the FRTB desk limits.

C1: Which Risk Model Would You Certify for a Romanian Bank's Trading Book? (2/2) [Proposed]

Solution discussed in class

Task

- (d) Rank the models with the FZ0 loss, DM tests with Newey–West variance and the 90% MCS.
- (e) Report the size and power of your test battery at $T = 250$, as in B1(e).
- **Deliverable:** a Quantlet folder in the team repository and one slide with the recommended model, the evidence table and the main limitation.
- **Answer:** would your choice survive a year like 2008 or 2020?

C2: Critique an AI Answer (1/3) [Proposed]

Solution discussed in class

The student asks an AI assistant

- "I have daily S&P 500 closes in a pandas Series P. Write Python for a rolling 250-day historical-simulation VaR 1%, count the exceptions and run the Kupiec and Christoffersen tests."
- "My last 250 days have 7 exceptions. What does that mean for the model?"

The AI answer: text

- Losses $L_t = -r_t$; VaR_t is minus the 1% quantile of the last 250 returns; exception: $L_t > \text{VaR}_t$
- `lr_ind(I)` returns the Christoffersen independence statistic LR_{ind} Christoffersen (1998)
- Your last 250 days: $x = 7$, $\text{LR}_{uc} = 5.50$, $p = 0.019$: correct coverage is rejected at 5%
- Basel traffic light: 7 exceptions is the red zone, so the model must be withdrawn BCBS (1996)

C2: Critique an AI Answer (2/3) [Proposed]

Solution discussed in class

The AI answer: the code that came with the text

```
import numpy as np, scipy.stats as stats
r = np.log(P).diff().dropna().values
W, p = 250, 0.01
VaR = np.array([-np.quantile(r[t-W:t+1], p)
                 for t in range(W, len(r))])
I = (-r[W:] > VaR).astype(int)
T, x = len(I), I.sum()
ph = x / T
LR_uc = -2*((T-x)*np.log(1-p) + x*np.log(p)
           - (T-x)*np.log(1-ph) - x*np.log(ph))
LR_cc = LR_uc + lr_ind(I)
p_uc = stats.chi2.sf(LR_uc, df=1)
p_cc = stats.chi2.sf(LR_cc, df=1)
```

C2: Critique an AI Answer (3/3) [Proposed]

Solution discussed in class

Task

- ▣ Find the three main errors in the answer, then check the evaluation window and numerical boundary cases
- ▣ For each error, say how you would detect it: a test, a published number, the lecture
- ▣ Give the corrected version: formula, code line and number
- ▣ Log each error as in `AI_ERRORS.md`
 - ▶ What you asked, what the AI answered, what was wrong, how you found out, the correct version

Hints

- ▣ Which returns are known at the close of day $t - 1$, when VaR_t is reported?
- ▣ How many restrictions does the conditional coverage test impose?
- ▣ Compare the zone with A5 and with the Basel table [BCBS \(1996\)](#)

C3: Can the Tail Data Rank These ES Forecasts? (Case Study) (1/2) [Proposed]

Solution discussed in class

Task

- ▣ **Context:** in the Lecture 8 case study [Pele & Mazurencu-Marinescu-Pele \(2026\)](#), a comparison of two ES forecasts is precision-fragile when $|\bar{r}_n^{(1)} - \bar{r}_n^{(2)}| < \sqrt{\hat{C}_1^2 + \hat{C}_2^2} / \sqrt{n\alpha}$; you audit the ES 2.5% rankings of Lecture 8 with this screen.
- ▣ (a) Take the VaR 2.5% and ES 2.5% forecasts of the six Lecture 8 models for the S&P 500, BET, Bitcoin and EUR/RON; on rolling 250-day windows moved forward by 21 days, estimate the additive corrections $(\hat{q}_n, \hat{r}_n)$ that minimise the mean FZ0 loss (Nelder–Mead with three random starts, as in the paper).
- ▣ (b) For each market and model, compute $\hat{C}$ as the standard deviation of $L_t - \text{VaR}_t$ on breach days, the floor $B = \hat{C} / \sqrt{n\alpha}$ with $n = 250$ and $\alpha = 2.5\%$, and the ratio R of the standard deviation of $\hat{r}_n$ (after removing its 252-day moving average) to B .
- ▣ (c) Classify the 15 model pairs of each market (60 pairs) with the screen, then repeat with the tolerance multiplied by $\kappa \in \{0.5, 1, 1.5, 2\}$; report the fragile share for each κ next to Table 14 of the paper (16.0%, 20.1%, 31.2%, 45.1%).

C3: Can the Tail Data Rank These ES Forecasts? (Case Study) (2/2) [Proposed]

Solution discussed in class

Task

- (d) Build out-of-sample FZ0 losses (correction estimated on a window, applied to the next 21 days), run DM tests [Diebold & Mariano \(1995\)](#) with Newey–West variance [Newey & West \(1987\)](#) and the [Harvey, Leybourne & Newbold \(1997\)](#) correction on all 60 pairs, and cross-tabulate them with the screen as in Table 15 of the paper.
- (e) Code the sample-size rule $n \geq f(n, \alpha)^2 (C/\varepsilon)^2 / \alpha$, $f(n, \alpha) = \sqrt{1 + (1 - \alpha)/(n\alpha)}$, by fixed-point iteration; check it on Table 2 ($\alpha = 2.5\%$, $C = 1.152$: 246, 627, 1 365, 5 348 days for $\varepsilon = 0.5, 0.3, 0.2, 0.1 C$), then give, for each market, the window needed for $\varepsilon = 50$ bp with your $\hat{C}$.
- **Deliverable:** a Quantlet folder in the team repository and two slides: the fragile shares with the DM cross-table, and the sample-size table by market.
- **Model:** B5 (Diebold–Mariano with HAC variance) and the case study of Lecture 8.
- **Answer:** which Lecture 8 ES rankings (FZ0, DM, MCS) would you not use as evidence for replacing a model, and why?

Key Takeaways

- ▣ Kupiec examines breach counts; Christoffersen, duration tests and DQ examine timing: always use both types of test; know their size and power at your T
- ▣ Estimated parameters with P/R large distort χ^2 critical values: bootstrap with re-estimation
- ▣ The traffic light is a Binomial test; it can let an inaccurate model through one time in eight
- ▣ Z_2 and PIT-based ES tests need the forecast distribution (simulate p -values); residual-bootstrap and regression tests need less
- ▣ Rank with FZ0 + DM (HAC) + MCS; significance needs many tail days
- ▣ Conformal coverage is marginal; in crises it fails like an unconditional model

Take-Home Assignment (Team Project)

- Choose a portfolio not covered in class (a BVB stock, a currency or a crypto-asset); forecast VaR 1%, VaR 2.5% and ES 2.5% with at least three models
- Backtest VaR 1% with Kupiec, Christoffersen and durations, and the pair (VaR 2.5%, ES 2.5%) with Z_2 ; place each model in the traffic light for its worst 250-day window
- Rank them with FZ0 on (VaR 2.5%, ES 2.5%), Diebold–Mariano tests (HAC) and the MCS
- Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- Log every AI error you catch (at least three per project) in `AI_ERRORS.md` and declare your AI use in `AI_USE.md`

References (1/3)

- ▣ Acerbi, C., Szekely, B. (2014). [Backtesting expected shortfall](#). MSCI Inc., December 2014.
- ▣ Basel Committee on Banking Supervision (1996). [Supervisory framework for the use of “backtesting” in conjunction with the internal models approach to market risk capital requirements](#). Bank for International Settlements.
- ▣ Basel Committee on Banking Supervision (2019). [Minimum capital requirements for market risk](#) (rev. February 2019). Bank for International Settlements.
- ▣ Christoffersen, P.F. (1998). [Evaluating interval forecasts](#). *International Economic Review*, 39(4), 841–862.
- ▣ Christoffersen, P., Pelletier, D. (2004). [Backtesting value-at-risk: a duration-based approach](#). *Journal of Financial Econometrics*, 2(1), 84–108.
- ▣ Diebold, F.X., Mariano, R.S. (1995). [Comparing predictive accuracy](#). *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Du, Z., Escanciano, J.C. (2017). [Backtesting expected shortfall: accounting for tail risk](#). *Management Science*, 63(4), 940–958.

References (2/3)

- ▣ Ehm, W., Gneiting, T., Jordan, A., Krüger, F. (2016). [Of quantiles and expectiles: consistent scoring functions, Choquet representations and forecast rankings](#). *Journal of the Royal Statistical Society, Series B*, 78(3), 505–562.
- ▣ Engle, R.F., Manganelli, S. (2004). [CAViaR: conditional autoregressive value at risk by regression quantiles](#). *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Escanciano, J.C., Olmo, J. (2010). [Backtesting parametric value-at-risk with estimation risk](#). *Journal of Business & Economic Statistics*, 28(1), 36–51.
- ▣ Fissler, T., Ziegel, J.F. (2016). [Higher order elicibility and Osband's principle](#). *Annals of Statistics*, 44(4).
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Giacomini, R., White, H. (2006). [Tests of conditional predictive ability](#). *Econometrica*, 74(6), 1545–1578.
- ▣ Gibbs, I., Candès, E. (2021). [Adaptive conformal inference under distribution shift](#). *Advances in Neural Information Processing Systems*, 34; arXiv:2106.00170.

References (3/3)

- ▣ Gneiting, T. (2011). [Making and evaluating point forecasts](#). *Journal of the American Statistical Association*, 106(494), 746–762.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). [The model confidence set](#). *Econometrica*, 79(2), 453–497.
- ▣ Harvey, D., Leybourne, S., Newbold, P. (1997). [Testing the equality of prediction mean squared errors](#). *International Journal of Forecasting*, 13(2), 281–291.
- ▣ Kupiec, P.H. (1995). [Techniques for verifying the accuracy of risk measurement models](#). *Journal of Derivatives*, 3(2), 73–84.
- ▣ McNeil, A.J., Frey, R. (2000). [Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach](#). *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ Newey, W.K., West, K.D. (1987). [A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix](#). *Econometrica*, 55(3), 703–708.
- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). [Dynamic semiparametric models for expected shortfall \(and value-at-risk\)](#). *Journal of Econometrics*, 211(2), 388–413.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). [Finite-sample precision limits for expected shortfall forecast comparisons](#). *Mathematics*, 14(13), 2316.